

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC

Subject: Physics

Questions: 40

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AI-generated PYQs: 0

1. [UNKNOWN] paper-i

1. (a) Construct the Hamiltonian of a charged particle with charge q and mass m moving with the velocity v in the external electromagnetic field, $\mathbf{E} = E_0 \hat{i}$, $\mathbf{B} = B_0 \hat{k}$, where E_0 and B_0 are constants. Calculate the first relativistic correction to the kinetic energy of a particle with rest mass m_0 and speed v . Calculate the focal lengths of the components of an achromatic telescope objective having a power of + 3.333 diopter, made from crown glass and flint glass, whose dispersive powers are 0.012 and 0.02 respectively. A particle moving in a central force field located at $r = 0$, describes a spiral, $r = et$ Find the force law. Thin transparent films produce colours when a white light falls on them or they act as antireflecting coatings. Find the conditions for these characteristics. 8

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2. [UNKNOWN] paper-i

Q2. (a) Discuss the mechanics of a system of point particles with special emphasis on the conservation theorems. How can we extend the results to a system with continuous mass distribution? 10 (i) State and prove Hamilton's principle and use it to prove that the shortest distance between two points in space is a straight line joining them. (ii) Use Hamiltonian mechanics to find the differential equation for planetary motion, moving under force $f(r) = -\frac{4}{r^3}$ and prove that r the areal velocity is constant. 8+7=15 (i) What is Holography? Show with simple diagrams, how a hologram is written and read using a laser. Mention some important applications of holography. 3+8+4=15 FSI-P-P/ir ===== PAGE 3 =====

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3. [UNKNOWN] paper-i

2. (a) A particle of mass m is constrained to move on a plane curve $xy = c$ with $c > 0$, under gravity, where y -axis is vertical. Construct the Lagrangian of the system and obtain the Euler-Lagrange equation of motion. 10 Consider a homogeneous cube of total mass M and side a . Taking the origin at one corner of the cube and axes along the edges of the cube, construct the moment of inertia tensor. Calculate principal moment of inertia. 15 (1) Give the equation of motions for a particle executing simple harmonic motion for undamped, damped and forced vibrations, and explain the terms. (ii) Give a plot between the displacement and time for each case of the above and interpret. (iii) What are the conditions for critical damping and resonance? 8+4+3=15

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4. [UNKNOWN] paper-i

Q3. (a) State the fundamental postulates of Einstein's special theory of relativity. Deduce Lorentz transformation equation and discuss how this accounts for the phenomenon of length contraction. 70 Discuss the properties of Cornu spiral. Show that the spiral can be used to obtain the intensity distribution in the Fresnel's diffraction pattern due to a straight edge. 10 (i) Using the concept of spontaneous and stimulated emission of radiation, obtain the relation between Einstein's A and B coefficients. What is the physical significance of Einstein's A coefficient? Justify why lasing action is much more difficult at X-ray frequency than in case of infrared frequency spectrum. 10+5+5=20

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5. [UNKNOWN] paper-i

3. (a) In special theory of relativity, the Hamiltonian of a free particle with rest mass m_0 is given by $\sqrt{p^2 c^2 + m_0^2 c^4}$. Obtain the Lagrangian of the system using Legendre transformation. 10 (b) Show that the wave equation for the propagation of electromagnetic scalar potential $\phi(x, y, z, t)$ $\nabla^2 \phi = \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2}$ remains invariant under Lorentz transformation. 10 8 8 8 8 FS1-P-PHY/48 3 [P.T.O. ===== PAGE 4 ===== • (c) (i) When a monochromatic light passes through a single slit, dark and bright fringes are observed on a screen kept far away from the slit. Draw an amplitude and intensity pattern of the light on the screen. Write an expression for intensity distribution and give the conditions for bright and dark fringes. If the slit width is 0.3 mm and the distance between the screen and the slit is

about 30 cm, then what is the angle of diffraction for the first minimum? ($\lambda = 5 \times 10^{-5}$ cm) 8+8+4=20

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6. [UNKNOWN] paper-i

Q4. (a) (i) In a Michelson's interferometer, 100 fringes cross the field of view when the movable mirror is displaced through 2.894×10^{-3} cm. Calculate the wavelength of the monochromatic source of light. (ii) A shift of 200 fringes is observed when the movable mirror of a Fabry-Perot interferometer is shifted by 0.0298 mm. Calculate the wavelength of the incident radiation. 8+7=15 (b) State and explain Fermat's principle of extremum path and use the same to deduce the laws of reflection and refraction of light. 10 (c) (i) Explain the reason for pulse broadening due to intermodal and material dispersion. Deduce the relation of pulse broadening for intermodal dispersion in optical fiber. (ii) A step index fiber in air has a numerical aperture of 0.16, a core refractive index of 1.45 and a core diameter of 60 μ m. Determine the normalized frequency for the fiber when light at a wavelength of 0.8 μ m is transmitted. Also estimate the number of guided modes propagating in the fiber. 10+5=15 3 ===== PAGE 4 ===== SECTION B

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7. [UNKNOWN] paper-i

4. (a) (i) What is population inversion? How is it efficiently achieved in He-Ne laser? Show it with necessary energy levels diagram. (ii) Show that for a normal optical source emitting at 600 nm at a temperature of 103 K, the spontaneous emission dominates over stimulated emission. 8+7=15 With the help of a neat diagram, explain spherical aberration. Briefly discuss the methods to minimize spherical aberration. 10 (1) How are linearly polarised and circularly polarised lights generated from unpolarised light and are analysed? (ii) A calcite crystal is to be used for producing a quarter wave plate. The refractive indices of the crystal for ordinary ray and extraordinary ray are 1.65836 and 1.48641 respectively. What is the thickness of the plate required if the light of wavelength 632.8 nm is to be used? 10+5=15 sEcrioN—B

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8. [UNKNOWN] paper-i

Q5. Answer the following: 8x5=40 In a one-dimensional device, the charge density is given by $\rho_v = \rho_0 - a \cdot x$. If $E = 0$ at $x = 0$ and $V = 0$ at $x = a$, find V and E using Laplace equation of electrostatics. 8 State and explain the Biot-Savart law. Derive an expression for the magnetic field at a point due to an infinitely long straight current carrying conductor. 8 Write down the four Maxwell's equations and explain the contribution of Maxwell in the development of these equations. 8 Prove the thermodynamic relation: $(\partial S / \partial P)_T = -(\partial V / \partial T)_P$ and hence show that $dP/dT = (V_2 - V_1) / (T_2 - T_1)$ all the terms have their usual meanings. 8 Describe neutron star on the basis of Fermi-Dirac statistics and obtain the condition of critical mass for a neutron star. 8

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9. [UNKNOWN] paper-i

5. (a) There is a potential gradient of 100 V/m normal to the surface of the earth. Assuming the earth to be a charged sphere of radius 6370 km, find the total charge on the earth. 8 State and explain Biot-Savart law. Obtain an expression for the magnetic field at the center of a circular loop of radius r metres, carrying a current of I amperes. 8 In free space, the electric field of electromagnetic wave is given by $(x, t) = 100 \cos(\omega t - kx)$ volt/metre. Find the average power crossing a circular area of radius 2 metres in the yz -plane. 8 FSI-P-PHY/48 4 ===== PAGE 5 ===== (d) Consider N molecules of a gas obeying van der Waals' equation of state given by $(P + a/V^2)(V - b) = Nk_B T$ where a is a measure of the attractive forces between the molecules and b is another constant proportional to the size of the molecules. The other symbols have their usual meanings. Show that during an isothermal expansion from volume V_1 to volume V_2 quasi-statically and reversibly, the work done is $W = -Nk_B T \ln(V_2 - Nb) + aN(1/V_2 - 1/V_1)$. (e) Starting from the first law of thermodynamics, show that $1/T = (\partial S / \partial U)_{V, N}$. 8

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10. [UNKNOWN] paper-i

Q6. (a) Use the method of electric images to find the electric field on the surface of a grounded conducting sphere. 10 (b) (i) State Faraday's law of electromagnetic induction and prove that it can be expressed in the following vector form: $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$ with $\mathbf{E}$ and $\mathbf{B}$ being the electric and magnetic fields. (ii) A coil of 10 turns has dimension 9 cm x 7 cm. It

rotates at the rate of 1571 rad/sec in a uniform field whose flux density is 0.6 weber/m². What is the maximum e.m.f. induced in the coil? 10+5=15 F.S./P-M-ty ===== PAGE 5 ===== (c) How does one explain the observed spectrum of black-body radiation using Planck's quantum hypothesis? State and obtain Wien's displacement law. Also explain the important features of this law. 15

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11. [UNKNOWN] paper-i

6. Obtain an expression for the electric potential and electric field strength at a point due to an electric dipole. 15 (1) Show that the electric and magnetic energy densities in a plane travelling wave are equal. Also prove that the total energy density = $60E^2 = 110112$. (ii) Deduce the equation of continuity based on Maxwell's equations. 10+5=15 Explain Planck's formula for black-body radiation. Use this formula to show that energy radiated per unit time per unit area in the range dX of A is $e(x, T)_{(271c^2h)} (ephc/X)_{[1 dx X.5 10}$

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12. [UNKNOWN] paper-i

Q7. (a) Using a c&t What of energy Maxwell's equations, obtain the relation $(E^2 \pm B^2) v. E B 0 + (x) = 2$ is Poynting vector? Deduce Poynting theorem for the flow in an electromagnetic field. 5+10=15 Discuss the reflection and refraction of plane electromagnetic waves at plane dielectric boundaries for normal incidence and also find the reflection and transmission coefficients. 15 What do you understand by spontaneous magnetization below Curie temperature? Explain with an appropriate diagram, the occurrence of a hysteresis loop in a ferromagnetic material. 10

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13. [UNKNOWN] paper-i

7. (a) Using the set of four Maxwell's equations, obtain the Lorentz condition relation between the scalar potential (1) and vector potential A , i.e., $\nabla \cdot A = -\frac{1}{c^2} \frac{\partial \rho}{\partial t}$ and discuss the gauge transformations, Lorentz gauge and Coulomb gauge. 15+5=20 FSI-P-PHY/48 5 [P.T.O. ===== PAGE 6 ===== Two resistors of 600 Ω and 800 Ω are connected in series with a 7 volts battery. An ammeter of 10 Ω resistance is used to measure current. N What will be the reading in the ammeter? (ii) Similarly if a voltmeter of 10000 Ω resistance is used to measure the potential difference across the 600 Ω resistor, what will be the reading in the voltmeter? 7 volts 7 volts 10000 Ω 4+6=10 Define Fermi energy and show that for an electron gas at absolute zero temperature, the Fermi energy is given by $E_F = \frac{h^2}{2m} (3\pi^2 n)^{2/3}$ where the symbols have their usual meanings. Estimate the numerical value of Fermi energy for copper taking number of electrons per unit volume as 8.4×10^{22} electrons/cm³ and find Fermi temperature T_F 10

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14. [UNKNOWN] paper-i

Q8. (a) State Maxwell's distribution law of molecular speeds. Draw and explain a curve between $n(c)$ and c in a gas at a given temperature T , where $n(c)dc$ is the number of molecules having speed between c and $c + dc$. Discuss the effect of T and mass m of the molecule on the nature of the curve. 15 (i) Define and explain the significance of the quality factor of an electrical machine. (ii) Discuss in brief, the working principle of a transformer. 5+5=10 Derive the mathematical expression for the total energy of a degenerate Fermi gas at a temperature T and calculate the specific heat of the Fermi gas at this temperature. 15

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15. [UNKNOWN] paper-i

8. (a) Write Bose-Einstein distribution function explaining every symbol. Derive Einstein's result on specific heat of solids explaining the assumptions made in the model. Discuss the low and high temperature limits of the predicted specific heat. In which limit, the Einstein formula fails to explain the experimental data? 10 (b) (1) The equation for an alternating current is $i = 42.42 \sin(314t)$. Find the following: Maximum value of current, Frequency, RMS value and Average value (ii) A condenser of capacity 1 μF is first charged and then discharged through a resistance of 1 $M\Omega$. Calculate the time in which the charge on the condenser will fall to 50% of its initial value. FSI-P-PIY/ 48 6 ===== PAGE 7 ===== (iii) Consider the displacement vector $\vec{r}$, given by $\vec{r} = (10xy^2z + 4x^2y)\hat{i} + (5xz^2 + 10x^2yz)\hat{j} + nCz\hat{k}$ Find the total charge enclosed in a cube of volume 10⁻⁹ m³ located at the point (1, 2, 3). 5+5+5=15 (c) Explain Maxwell-Boltzmann formula

for distribution of velocities of gas molecules at temperature T . What will be the formula for distribution of speeds? If P , v , v_{rms} , and v_{mp} denote average speed, root mean square velocity and most probable speed, show that : $v : v_{rms} : v_{mp} = 1 : \sqrt{3} : 1$ *
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16. [UNKNOWN] paper-ii

0. Find the energy E corresponding to the eigen state given by $\psi(x)$. Is it a ground state energy or one of the excited state energies ? Comment. (b) Consider an experiment in which a beam of electrons is directed at a plate containing two slits, labelled as A and B. Beyond the plate is a screen, where electrons hit the screen and are detected. For each of the following cases sketch the variation of the relative number of incident electrons as a function of position along the screen and also provide brief explanation about each observation: Slit A open, slit B closed, Both A and B are open. ES-13-7,H-r 4 15 20 10 ===== PAGE 5 ===== (c) Consider a particle trapped in a box of width L and its n th state wave function is given by $\psi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$. Calculate the expectation value of the position $\langle x \rangle$ of the particle. How is this result different from classical consideration of finding the particle inside the box? 10

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17. [UNKNOWN] paper-ii

1. (a) Use the uncertainty principle to estimate the binding energy of the hydrogen atom in the ground state in terms of m_e , e , h and c . Estimate the answer in eV. 8 I. (b) An electron is described by the following wave function: $\psi(x) = C e^{-x/a}$ for $x \geq 0$ and $\psi(x) = 0$ for $x < 0$ where C is a constant. Find out the average position of the electron. 8

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18. [UNKNOWN] paper-ii

1. (c) What is the main inference drawn from the Stern Gerlach experiment ? What would be the change in the result in the following cases : _electron spin is $-\frac{1}{2}$ instead of a beam of neutral atoms is passed through a homogeneous magnetic field. 8

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19. [UNKNOWN] paper-ii

1. (d) What is meant by Larmour precession? Calculate the Larmour frequency for protons in a magnetic field of 1 Tesla. 8 (e) Describe the differences in the behaviour of a quantum harmonic oscillator with the corresponding classical oscillator. 8 (a) A particle of mass m is confined in a one dimensional box on the x -axis between $x = 0$ and $x = L$. Obtain the allowed energy states of the particle and the corresponding eigenfunctions. What will be the energy pattern in the limit of $L \rightarrow \infty$? 15

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20. [UNKNOWN] paper-ii

2. (b) Estimate the probability of finding the particle between $x=0$ and $x = -L$ in the 3 lowest energy state. 5

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21. [UNKNOWN] paper-ii

2. (c) Generalize the results obtained in part (a) to obtain the allowed energy states and the corresponding eigenfunctions of a particle in a three dimensional isotropic box of dimension L . What is the degeneracy of the first two excited states of the particle? 20

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22. [UNKNOWN] paper-ii

Q2. (a) How many different wavelengths may be observed in the spectrum from a hydrogen sample, if the atoms are excited to states with principle quantum number n ? 5 FSI-D -m-ty 3 ===== PAGE 4 ===== (b) The radial part of Schrodinger wave function for hydrogen atom for e^2 spherical symmetric potential $V(r) = -\frac{e^2}{4\pi\epsilon_0 r}$ is given as: $R_{nl}(r) = \frac{2}{\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} e^{-\rho} \rho^L L_n^{2L}(\rho)$ where $\rho = \frac{2Zr}{a_0}$, a_0 is Bohr radius, Z is atomic number, L is angular momentum quantum number, n is principal quantum number, m is mass of electron, M is mass of proton respectively. Obtain the form of the above equation for ground state electron of hydrogen atom. Also starting

with a trial wave function for the radial equation, $R = A e^{-r/a_0}$, for the $l = 0$ state, find the expressions of energy E and orbital radius for the ground state of hydrogen atom. 5+15=20 (c) The quantum theory describes the angular momentum (J) as a linear operator with three components J_x , J_y and J_z in three dimensions. For the angular momentum operator prove that $[J^2, J_i] = 0$, where $i = x, y, \text{ and } z$ and comment on the measurability of J^2 and its components as operators.

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23. [UNKNOWN] paper-ii

3. (a) Consider two electrons system in $(3p)(4s)$ configuration. Derive the allowed energy states of the system in (i) LS coupling and (ii) jj coupling sketching the energy level diagram in each case. Name the interactions which are responsible for different splittings. 25

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24. [UNKNOWN] paper-ii

3. (b) Show that (i) $\nabla \times \nabla \phi = -\nabla^2 \phi$, (iii) $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ & $\nabla \times (\nabla \phi) = 0$ fsi-d-phy 2 15 ===== PAGE 3 ===== 4.(a) A beam of particles of mass m and energy E is incident on a step potential of height V_0 from the left as shown in the figure. Discuss the behaviour of the particle for (i) $E < V_0$ and (ii) $E > V_0$. Obtain expressions for the reflection and the transmission coefficients and sketch them as a function of the incident energy of the particle. 15

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25. [UNKNOWN] paper-ii

Q3. (a) Write down the Schrodinger wave equation for a one-dimensional harmonic oscillator in which a particle of mass m and frequency ω is subject to a parabolic potential $V(x) = \frac{1}{2} m \omega^2 x^2$. Let one of the possible energy eigen states be given by $\psi(x) = A e^{-\alpha x^2}$

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26. [UNKNOWN] paper-ii

4. (b) The first rotational line in the rotational spectrum of CO is observed at 3.84235 cm^{-1} . Calculate the rotational constant B and the bond length. 10

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27. [UNKNOWN] paper-ii

4. (c) What are Stokes and anti-Stokes lines? What is their importance in understanding the molecular structure ? 15
SECTION 'B'

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28. [UNKNOWN] paper-ii

Q4. (a) Show that for pure rotational spectrum of a diatomic molecule the wave number difference between the consecutive lines is $2B$, where B is the moment of inertia. 20 If $\nu = 4050 \text{ m}^{-1}$ for the HF molecule, what is the interatomic distance of this molecule? [Given $I = 1.05 \times 10^{-40} \text{ Js}$] 10 The most intense vibrational bands of CO and HCN molecules have the wave numbers $2143 \times 10^3 \text{ m}^{-1}$ and $2.886 \times 10^5 \text{ m}^{-1}$ respectively. Calculate the force constants of these molecules. 10 FSI-A-FT-fr 5 ===== PAGE 6 ===== SECTION B

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29. [UNKNOWN] paper-ii

5. Answer all of the following questions: 8x5=40

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30. [UNKNOWN] paper-ii

5. (a) How much energy is required to separate a nucleus into its constituent nucleons? What is the binding energy per nucleon for this nucleus? 8 5.(b)

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31. [UNKNOWN] paper-ii

5. (Determine the values of emitter current and the collector current of the transistor having $\alpha = 0.98$ and collector to base leakage current $I_{mo} = 4 \mu A$. The base current is $50 \mu A$. 8 What are noncentral nuclear forces? Discuss qualitatively how the presence of a small noncentral component in the nuclear force can account for the nonvanishing quadrupole moment of deuteron. 8 Define the terms critical magnetic field and critical temperature in a superconductor. Find the magnetic field strength necessary to destroy the superconductivity in a sample of lead at 4.2 K. The critical magnetic field at 0 K is 0.80 Tesla and the critical temperature is 7.2 K. 8 Write the quark contents and the quantum numbers (B, 1, 5, 13) of the following hadrons : K^0 8 3 fsi-d-phy ===== PAGE 4 =====

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32. [UNKNOWN] paper-ii

Q5. (a) The nuclear radius of $^{80}_{16}O$ is 3 fermi. What will be the nuclear radius of $^{235}_{92}U$? 8 Draw the baryon octet according to Murray Gell-Mann. 8 Show that the probability of occupation for an electron state at the Fermi energy is equal to 0.5 for all finite temperatures. 8 A small magnet is placed on the surface of a disc of normal material. Explain what happens when the disc is cooled down so that it becomes superconducting. Can this be explained from classical viewpoint if we simply treat the superconductor as a material which has zero resistivity? 8 Find the output voltage of the three-input adder circuit shown below. 8

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33. [UNKNOWN] paper-ii

6. (a) What is meant by quark confinement? Why are the masses of hadrons larger than the sum of the masses of their constituent quarks ? 12

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34. [UNKNOWN] paper-ii

6. (b) Describe the main features of the shell model of the nucleus. Explain how it can account for the magic numbers of nuclei. - 20 (c) Predict the spin and parity for the ground state of the following nuclei on the basis of the shell model : $^{15}_7N$ $^{16}_8O$ $^{17}_8O$ $^{18}_8O$ $^{8}_4Be$ $^{8}_4He$. / (a) Write down the semiempirical mass formula for nuclei and sketch the variation of isobaric masses with atomic number for both odd and even values of A. How many stable isobars would be there in each case ? 20 7. (b) What is Meissner effect? Explain perfect diamagnetism in reference to this. 10

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35. [UNKNOWN] paper-ii

Q6. (a) What is the source of the Sun's energy ? Explain with proper nuclear reaction. /0 What do you mean by nuclear reactors ? With a neat diagram, describe the essential parts of a nuclear reactor and discuss about its applications. 20 An atomic power reactor can deliver 300 MW. If due to fission of each atom of $^{235}_{92}U$, the energy released is 200 MeV, calculate the mass of uranium fissioned per hour. 10 6 ===== PAGE 7 =====

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36. [UNKNOWN] paper-ii

7. (c) Explain briefly SU(3) flavor symmetry and SU(2) isospin symmetry in particle physics. 10

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37. [UNKNOWN] paper-ii

Q7. (a) Explain drift velocity, mobility and conductivity of an intrinsic semiconductor. 5 Calculate the total conductivity in an intrinsic semiconductor. How does this conductivity change with temperature? 15 Derive the expression for electron concentration in the conduction band of an intrinsic semiconductor. 20

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38. [UNKNOWN] paper-ii

8. (a) Explain with the help of a neat diagram the working of an RC coupled common emitter amplifier. 15 8. (b) Find the Boolean algebra expression for the above system. 15

39. [UNKNOWN] paper-ii

8. (c) Why does the resistance of a semiconductor decrease with increase in temperature? 10 fsi-d-phy

40. [UNKNOWN] paper-ii

Q8. (a) Draw the circuit of a phase shift oscillator using transistor. Obtain the expression for the frequency of oscillations and condition for sustained oscillations. Give necessary equivalent circuit while deriving the expression of frequency of oscillations and condition for sustained oscillations. 20 A pair of transistors operates in a class B push-pull circuit with a power supply of voltage $V_{cc} = 12\text{ V}$. If the turns ratio of the output transformer is 2 and the load resistance R_L is $6\ \Omega$, calculate the maximum output power and maximum dc power supplied. Hence find efficiency of the circuit. Neglect transformer winding resistance. 10 Estimate the current gain of an n-p-n transistor if only 0.3% of the conduction electrons that enter the base from the emitter recombine in the base region. 10 FSI-D-Pfly